

JOHN CULNAN

jculnan@gmail.com

EDUCATION

- PhD in Linguistics**, *University of Arizona* Dec 2021
Dissertation: Acoustic Analysis and Automatic Classification of Personality and Individual Expression from Speech
- MS in Human Language Technology**, *University of Arizona* May 2019
- MA in Linguistics (distinction)**, *The University of Hong Kong* Dec 2016
Thesis: On the features of Hong Kong English: A corpus analysis
- BA in Linguistics & Spanish**, *Emory University* May 2012
minor coursework completed in French

EXPERIENCE

- NLP Data Scientist / NLP Team Lead** Mar 2024–Present
Massachusetts Veterans Epidemiology Research and Information Collaborative (MAVERIC), VA Boston
- Lead and support NLP algorithm development efforts for extraction of medical data elements from the electronic health record within the VA infrastructure using large language models and rule-based models.
 - Research and write manuscripts for publication based on our NLP and epidemiological studies.
 - Lead NLP team efforts to develop infrastructure and best practices for medical NLP research within the Boston Informatics Group at MAVERIC, including taking ownership of codebases for the creation of NLP algorithms, mentorship, and coordination of duties within the NLP team
 - Participate in the preparation of grant materials and pilot work to be used during grant submission for NLP-related research projects using VA infrastructure and data.
- Data Scientist** Dec 2023–Mar 2024
Boston Medical Center [Affiliated with MAVERIC at VA Boston during this time]
- Performed preprocessing, code development, and performing data analysis for a longitudinal study of multiple myeloma outcomes
 - Researched and wrote methodology and analysis of data for publication
 - Worked with the NLP team within the Boston Informatics Group at MAVERIC by developing a pilot transformer-based model of PD-L1 information extraction from medical notes
- Postdoctoral Research Associate I** Jan 2022–Mar 2023
University of Arizona
- Built, tested, and ran multimodal multitask networks for making predictions in a longitudinal behavioral health intervention (LiVES) and for AI supporting search and rescue teams (DARPA ASIST-ToMCAT)

- Designed and built data preprocessing pipeline for speech and text datasets
- Advised graduate students in speech- and multimodal natural language processing projects
- Developed in-depth reports that would be used in presentations with leaders, specifically reports on model creation, performance and results that were used in DARPA ASIST PI meetings as progress reports and to argue for continued model development and project funding
- Published results of NLP research using transformers for prediction of changes in mood and for high-quality adversarial data collection (see publications section)

Graduate Research Associate/Bilinski Fellow

Jan 2019–Dec 2021

University of Arizona

- normalfont Created multimodal multitask neural network models of speaker characteristics in Pytorch for DARPA ASIST-ToMCAT and the LIvES project under supervision of Dr. Adarsh Pyarelal, Dr. Rebecca Sharp, and Dr. Steven Bethard
- Prepared a baseline framework for multimodal metalearning in Pytorch
- Developed rule-based framework and scala testing framework for automatic extraction of music terminology for the Musica project
- Published results of work on emotion, sarcasm and personality detection, as well as science-domain corpus for entity recognition (see publications section)

Volunteer, CLU Lab

Sep–Dec 2018

University of Arizona

- Assisted in the translation of dependency grammars from Stanford to Universal Dependencies
- Participated in the creation of rules for rule-based information extraction system for biomedical entities

Graduate Intern

May–Sep 2018

Kamusi Project

- Updated Kamusi’s system of multi-word expression (MWE) extraction with NLTK in Python
- Analyzed existing framework of MWE extraction and recommended policy changes and improvements for immediate implementation, demonstrating skill in debate of policy issues affecting Kamusi’s program
- Developed an in-depth report of activities completed during the internship which was presented directly to the head of Kamusi to use for information on current system implementations and recommendations for future changes.

Graduate Research Assistant

Aug–Dec 2018

University of Arizona

- Worked as part of the development team for the Language Science and Innovation Collaborative (LSIC) under Dr. Sonia Shiri
- Researched and prepared applications for national grants to help fund language science research at the University of Arizona through the collaborative
- Co-organized and attended steering committee meetings and the Fall 2018 LSIC retreat

Graduate Teaching Assistant

Aug 2016–Dec 2018

University of Arizona

- Taught online courses in Language development (Fall 2017, Winter 2017, Spring 2018) and in-person sections for Language in the world (2016-2017)

- Prepared course materials and TAed for in-person courses in Phonetics (Fall 2018) and Language Development (Spring 2017)

Visiting Assistant Lecturer

Aug 2013–Jun 2015

Community College of City University, Hong Kong

- Taught English for Academic Purposes, English Professional Communication Skills, and Public Relations in Business and Society to 150-200 students per semester
- Developed materials and created examinations for Academic English and Effective English I
- Designed and organized activities to foster student development as part of a team
- Led small group, optional English sessions to 40-50 students per semester

English Teaching Assistant

Sep 2012–Jun 2013

Fulbright US Student Program, Macau

- Organized, prepared materials for and team-taught English conversation courses to 100+ undergraduates per semester at the University of Saint Joseph
- Designed and taught supplementary English writing classes for struggling third and fourth year students, focusing on the construction of better academic papers, from prewriting to editing

SKILLS

Natural Language Processing and Machine Learning: Large language models with LangChain, Langserve and Llama-cpp-python, text- and speech-based neural networks, multimodal modelling using FFNN, CNN, LSTM, and transformer (BERT-based) models in pytorch; rule-based and regex models

Dataset creation: Annotation of single-modality and multimodal natural language data, crowdsourcing responses, dataset cleaning, validation, organization, and quality control

Quantitative methods: statistical linear and logistic regression, classification, clustering, correlation, ANOVA, error analysis

Communication: Presentation of problems, data, and analysis to domain experts and non-experts, supervision of undergraduates, graduates and researchers, interdisciplinary collaboration

TECHNICAL KNOWLEDGE

Programming Languages

Python, R

NLP and ML libraries

Pytorch, pandas, Llama-cpp-python, SpaCy, HuggingFace, sklearn

Other technical skills

Label Studio, Prodigy, PyCharm, Git, Jupyter Notebooks, L^AT_EX

PUBLICATIONS

Lazzaro, A., Nannini, K., Yildirim, C., Huhmann, L., **Culnan, J.M.**, Corrigan, J.K., Elbers, D., Do, N.V., Brophy, M., Munshi, N., Fillmore, N.R., and Edwards, C.V. 2026. Evolution of Real-World Front-Line Treatment Patterns in Multiple Myeloma. Blood Advances. <https://www.sciencedirect.com/science/article/pii/S2473952926005379>

Cooperberg, M.R., Bihn, J.R., **Culnan, J.M.**, La, J., Goryachev, S.D., Chen, D.C., Soloviev, O., Lee, G., Corrigan, J.K., Swinnerton, K.N., Nickols, N.G., Dulberger, K.N., Barata, P., Bitting, R.L., Brophy, M.T., Cheng, H.H., De Hoedt, A., Do, N.V., Freedland, S.J., Garraway, I.P., Gaziano, J.M., Halabi, S., Haughey, R.L., Loeb, S., Nanus, D.M., Rebbeck, T.R., Rettig, M.B.,

Pan, C.X., Myrie, K., Ramoni, R.B., Fillmore, N.R., and Paller, C.J. 2026. Magnetic Resonance Imaging or Confirmatory Biopsy for Patients With Prostate Cancer Receiving Active Surveillance. *JAMA oncology*, 12(1), pp.111-113. <https://jamanetwork.com/journals/jamaoncology/article-abstract/2841675>

Barata, P.C., Corrigan, J.K., La, J., **Culnan, J.M.**, Akama-Garren, E., Dulberger, K.N., Dumontier, C., Hansen, J., Bihn, J.R., Bitting, R.L., Brophy, M.T., Cheng, H.H., Cooperberg, M.R., Do, N.V., Dorff, T., Fojo, A.T., Gaziano, J.M., Goryachev, S.D., Halabi, S., Haughey, R.L., Nanus, D.M., Rebbeck, T.R., Pan, C.X., Schoen, M.W., Swinnerton, K.N., Myrie, K., Ramoni, R.R., Fillmore, N.R., Paller, C.J., and Rettig, M.B. 2026. Docetaxel rechallenge vs cabazitaxel in patients with metastatic castration-resistant prostate cancer. *JAMA Network Open*, 9(1), p.e2551231. <https://jamanetwork.com/journals/jamanetworkopen/article-abstract/2844039>

Culnan, J.M., Goryachev, S.D., Bihn, J.R., Lee, G., Chen, D.C., Soloviev, O., Zwolinski, R., Dulberger, K.N., Do, N.V., Paller, C.J., Cooperberg, M.R., and Fillmore, N.R. 2025. Automated extraction of imaging and pathology data from diverse prostate cancer electronic records. *JCO Clinical Cancer Informatics*, 9, p.e2500085. <https://ascopubs.org/doi/full/10.1200/CCI-25-00085>

Pyarelal, A., **Culnan, J.M.**, Qamar, A., Krishnaswamy, M., Wang, Y., Jeong, C., Chen, C., Miah, M.M.M., Hormozi, S., Tong, J. and Huang, R., 2025, April. MultiCAT: multimodal communication annotations for teams. In *Findings of the Association for Computational Linguistics: NAACL 2025* (pp. 1077-1111). <https://aclanthology.org/2025.findings-naacl.61.pdf>

Culnan, J. and Yiu, S.Y. 2022. Distinct acoustic properties pattern differently in a sound change in progress in Hong Kong Cantonese. *Journal of the International Phonetic Association*. <https://doi.org/10.1017/S0025100322000172>

Culnan, J., Diaz, D.R. and Bethard, S., 2022, July. Exploring transformers and time lag features for predicting changes in mood over time. In *Proceedings of the Eighth Workshop on Computational Linguistics and Clinical Psychology* (pp. 226-231). <https://aclanthology.org/2022.clpsych-1.21.pdf>

Diaz, D.Y.R., Anioł, M. and **Culnan, J.**, 2022, July. Collecting high-quality adversarial data for machine reading comprehension tasks with humans and models in the loop. In *Proceedings of the First Workshop on Dynamic Adversarial Data Collection* (pp. 53-60). <https://arxiv.org/pdf/2206.14272.pdf>

Culnan, J., Park, S., Krishnaswamy, M. and Sharp, R., 2021, April. Me, myself, and ire: Effects of automatic transcription quality on emotion, sarcasm, and personality detection. In *Proceedings of the Eleventh Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis* (pp. 250-256). <https://aclanthology.org/2021.wassa-1.26/>

Smith, H., Zhang, Z., **Culnan, J.** and Jansen, P., 2020, May. ScienceExamCER: A high-density fine-grained science-domain corpus for common entity recognition. In *Proceedings of the Twelfth Language Resources and Evaluation Conference* (pp. 4529-4546). <https://www.aclweb.org/anthology/2020.lrec-1.558>

RECENT PROFESSIONAL CONFERENCE PRESENTATIONS

Culnan J.M., Ferri, G.F., La, J., DuMontier, C., Brophy, M.T., Do, N.V., Munshi, N.C., Edwards, C.V., and Fillmore, N.R. 2026. Identifying high-risk cytogenetic subgroups in multiple myeloma using large language artificial intelligence models. *2026 ASCO Annual Meeting*, Chicago, IL.

Corrigan, J.K., Paller, C.J., Fillmore, N.R., **Culnan, J.**, Dorff, T.B., Dulberger, K., Goryachev, S., Myrie, K., Pan, C., Rettig, M., Goodman, M.G., Barata, P., and Bitting, R.L. 2026. Sequencing of PARP inhibitors and taxane chemotherapy in HRR-altered metastatic castration-resistant prostate cancer: A real-world VA analysis.

Abdallah, M., Corrigan, J., **Culnan, J.**, Elbers, D., Eze-nwosu, C., Bell, B., Do N., Munshi, N., DuMontier, C., Fillmore, N., and Edwards, C. 2025. Chronological age predicts suboptimal multiple myeloma treatment independent of frailty and socioenvironmental factors: A retrospective multi-center cohort study. *The 67th ASH Annual Meeting*, San Diego, CA.

Lazzaro, A., Nannini, K., Yildirim, C., Huhmann, L., **Culnan, J.**, Corrigan, J., Elbers, D., Do, N., Brophy, M., Munshi, N., Fillmore, N., and Edwards, C. 2025. Real-world outcomes of newly diagnosed multiple myeloma patients in the veterans affairs healthcare system. *The 67th ASH Annual Meeting*, San Diego, CA.

Sommers, S., Zheng, C., Corrigan, J., **Culnan, J.**, Bodanapu, G., Ferri, G., Bell, B., Yellanki, M., Elbers, D., Do, N., Brophy, M., Munshi, N., Edwards, C., and Fillmore, N. 2025. Artificial intelligence/machine learning-based risk prediction model guiding individualized interventions to improve outcomes for patients with multiple myeloma. *The 67th ASH Annual Meeting*, San Diego, CA.

Wu, J.T., Conover, S., Su, C., Corrigan, J., **Culnan, J.**, Liu, Y., Kelley, M.J., Do, N., Arya, S., Harris, A.H.S., Langlotz, C., Wiener, R.S., Branch-Elliman, W., Han, S., and Fillmore, N. 2025. Enabling national identification of lung cancer screening eligibility with large language models. *2025 ASCO Annual Meeting I*, Chicago, IL.

Culnan, J., Goryachev, S., Chen, D., Soloviev, O., Lee, G., Bihn, J., Corrigan, J., Dulberger, K., La, J., Swinnerton, K., Dorff, T.B., Garraway, I., Halabi, S., Loeb, S., Nickols, N.G., Rettig, M., Schoen, M.W., Paller, C.J., Fillmore, N., and Cooperberg, M.R. 2025. A natural language processing algorithm to extract MRI and biopsy-related phenotypes for prostate cancer patients. *2025 ASCO Genitourinary Cancers Symposium*, San Francisco, CA.

Cooperberg, M.R., Bihn, J., **Culnan, J.**, La, J., Goryachev, S., Corrigan, J., Dulberger, K., Swinnerton, K., Lee, G., Garraway, I., Halabi, S., Loeb, S., Dorff, T.B., Hauger, R., Nickols, N.G., Rebbeck, T., Rettig, M., Schoen, M.W., Fillmore, N., and Paller, C.J. 2025. Can magnetic resonance imaging safely replace a confirmatory biopsy in patients on active surveillance for prostate cancer? *2025 ASCO Genitourinary Cancers Symposium*, San Francisco, CA.

Barata, P.C., Corrigan, J., **Culnan, J.**, Dulberger, K.N., La, J., Akama-Garren, E., DuMontier, C., Hansen, J.E., Bitting, R.L., Cheng, H.H., Cooperberg, M.R., Dorff, T.B., Halabi, S., Loeb, S., Nanus, D.M., Pan, C., Schoen, M.W., Fillmore, N., Paller, C.J., and Rettig, M. 2025. Docetaxel rechallenge versus cabazitaxel in patients previously treated with docetaxel for metastatic castrate-resistant prostate cancer (mCRPC). *2025 ASCO Genitourinary Cancers Symposium*, San Francisco, CA.

Srinivasan, M.D., **Culnan, J.**, Hora, M., Lazzaro, A., Bell, B., Bodanapu, G., Yildirim, C., Corrigan, J., Do, N.V., Brophy, M.T., Munshi, N.C., Szalat, R.E., Fillmore, N.R., and Edwards, C.V. 2024. Evaluation of socioeconomic barriers impacting patients with newly diagnosed multiple myeloma. *The 67th ASH Annual Meeting*, San Diego, CA.

Bodanapu G., Corrigan J., **Culnan, J.**, Srinivasan, M., Hora, M., Lazzaro, A., Yildirim, C., Elbers, D., Lang, D., Do, N.V., Brophy M.T., Bell, B., Szalat, R.E., Munshi, N.C., Fillmore, N.R.,

and Edwards, C.V. 2024. Risk factors for suboptimal treatment in patients with newly diagnosed multiple myeloma across US healthcare systems. *The 67th ASH Annual Meeting*, San Diego, CA.

Freylersythe, S., Sharp, R., **Culnan, J.**, Romero Diaz, D.Y., Zhao, Y., Franks. G.H., Nitschke. R., Bethard. S.J., and Crane. T.E. 2022. Lessons Learned from a Secondary Analysis Using Natural Language Processing and Machine Learning from a Lifestyle Intervention. *43rd annual meeting of the Society of Behavioral Medicine*.

Park, S. and **Culnan, J.**. 2021. Automatic proficiency judgments: accentedness, fluency, and comprehensibility. *181st Meeting of the Acoustical Society of America*. Seattle, WA.

Culnan, J., Crane, T.E., Sharp, R., Wright, S.J., Wang, T.T., Franks, G.H., Klimowski, C., Merchant, N. and Bethard, S.J.. 2020. Using natural language processing to determine predictors of healthy diet and physical activity behavior change in ovarian cancer survivors. *American Society of Preventive Oncology 2020*. Virtual.

Culnan, J., and Yiu, S.Y. 2019. Perceptual cues to a merger in progress. *177th Meeting of the Acoustical Society of America*. Louisville, KY.

Park, S. and **Culnan, J.**. 2019. A comparison between native and non-native speech for automatic speech recognition. *177th Meeting of the Acoustical Society of America*. Louisville, KY.

Culnan, J. and Yiu, S.Y. 2018. Acoustic features of labialized velars in a merger in progress. *175th Meeting of the Acoustical Society of America*. Minneapolis, MN.

Culnan, J. and Yiu S.Y. 2017. Measures of labialized and plain velars in Hong Kong Cantonese. *UA-ASU Cognitive Science Conclave*. Tucson, AZ.

AWARDS AND CERTIFICATIONS

Bilinski Foundation Graduate Fellowship (\$33,000)	2020-2021
SBSRI Pre-Doctoral Dissertation Grant (\$497)	2019
GPSC Travel Grant Recipient (\$496)	Spring 2019
GPSC Travel Grant Recipient (\$360)	Spring 2018
GPSC Travel Grant Recipient (\$330)	Spring 2017
Fulbright ETA Recipient, Macau	2012-2013